China — EDA, TTC-модели, SHAP

Что тут: EDA → Stage-1 (Through-The-Cycle) → проверка H1 и H2 по Китаю.

- **H1.** Ансамбли (RF, XGBoost) выдают более высокий ROC-AUC, чем Logit, но у них больше **зазор train–test** (переобучение).
- **H2.** Сумма `|SHAP|` по блокам **Ликвидность + Инновации (HMA+R&D)** выше, чем по блоку **Леверидж**.

Выборка: 339 активных Wind-компаний + 11 реально-дефолтных (11 vs 3412 строк, дисбаланс ≈ 1:71). Выбросы по делистингу с экономически нерелевантными причинами (M&A, приватизация) уже удалены на этапе загрузки (`30_china_load_and_clean.ipynb`) — то есть здесь мы проверяем Hypotheses именно на *true economic distress*.

Правила научрука: без SMOTE, `class_weight='balanced'`, stratified split. Артефакты сохраняются в `reports/china/`.

In [1]:
```
import warnings
warnings.filterwarnings('ignore')

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from pathlib import Path

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.metrics import roc_auc_score, average_precision_score

from xgboost import XGBClassifier
import shap

RNG = 42
sns.set_theme(style='whitegrid', context='notebook')
plt.rcParams['figure.dpi'] = 110

PROCESSED = Path('../data/processed')
REPORTS = Path('../reports/china')
REPORTS.mkdir(parents=True, exist_ok=True)

def save_fig(fig, name):
    path = REPORTS / f'{name}.png'
    fig.savefig(path, dpi=160, bbox_inches='tight')
    print(f'  saved → {path}')

_report_lines = ['# China — Stage-1 (TTC) results\n']
def log(msg=''):
    print(msg)
```

```
    _report_lines.append(msg)

df = pd.read_csv(PROCESSED / 'cn_panel_cleaned.csv', encoding='utf-8-sig')
print('Shape:', df.shape)
print('Tickers per target:', df.groupby('target')['ticker'].nunique().to_dict())
df.head(3)
```

```
Shape: (3460, 20)
Tickers per target: {0: 339, 1: 11}
```

Out[1]:

	ticker	company_name	year	total_revenue	ebit	ebitda	total_assets	total_lia
0	CCIHY.OO	CHINACACHE	2014	226.23	-2.86	0.0	282.92	146.4
1	CCIHY.OO	CHINACACHE	2015	208.46	-20.46	0.0	403.26	292.1
2	CCIHY.OO	CHINACACHE	2016	151.97	-70.09	0.0	256.14	276.6

1. Feature engineering

Мэппинг колонок `cn_panel_cleaned.csv` в ratios H2. В китайской выборке есть **R&D expense** — это отдельный канал «Innovation» сверх нематериальных активов, что важно для H2.

- **Liquidity:** `current_ratio`, `cash_to_assets`, `cash_to_cl`, `wc_to_assets`
- **Innovation:** `intangibles_to_assets`, `rd_to_revenue`
- **Leverage:** `debt_to_assets`, `debt_to_equity`, `lt_debt_to_assets`, `interest_coverage`
- **Profitability:** `roa`, `net_margin`, `operating_margin`, `cfo_to_assets`
- **Size:** `log_assets`, `log_revenue`

Винзоризация 1/99% — как в российской тетради. Пропуски заполняются медианой (после ffill/bfill внутри компании в loader-ноутбуке).

In [2]:

```
TARGET = 'target'
ID_COL = 'ticker'

A, CA, CASH, INT_A = 'total_assets', 'current_assets', 'cash', 'intangibles'
EQ, LIAB, ST_L = 'total_equity', 'total_liab', 'current_liab'
REV, EBIT, NI = 'total_revenue', 'ebit', 'net_profit'
INTEREST, CFO, RD = 'interest_expense', 'cfo', 'rd_expense'

def safe_div(a, b):
    b = b.replace(0, np.nan)
    return a / b

panel = df.copy()
lt_liab = (panel[LIAB] - panel[ST_L]).clip(lower=0)

panel['current_ratio']         = safe_div(panel[CA], panel[ST_L])
panel['cash_to_assets']        = safe_div(panel[CASH], panel[A])
panel['cash_to_cl']            = safe_div(panel[CASH], panel[ST_L])
panel['wc_to_assets']          = safe_div(panel[CA] - panel[ST_L], panel[A])

panel['intangibles_to_assets'] = safe_div(panel[INT_A], panel[A])
panel['rd_to_revenue']         = safe_div(panel[RD], panel[REV])
```

```
panel['debt_to_assets']        = safe_div(panel[LIAB], panel[A])
panel['debt_to_equity']        = safe_div(panel[LIAB], panel[EQ])
panel['lt_debt_to_assets']     = safe_div(lt_liab, panel[A])
panel['interest_coverage']     = safe_div(panel[EBIT], panel[INTEREST])

panel['roa']              = safe_div(panel[NI], panel[A])
panel['net_margin']       = safe_div(panel[NI], panel[REV])
panel['operating_margin'] = safe_div(panel[EBIT], panel[REV])
panel['cfo_to_assets']    = safe_div(panel[CFO], panel[A])

panel['log_assets']  = np.log1p(panel[A].clip(lower=0))
panel['log_revenue'] = np.log1p(panel[REV].clip(lower=0))

FEATURES = [
    'current_ratio', 'cash_to_assets', 'cash_to_cl', 'wc_to_assets',
    'intangibles_to_assets', 'rd_to_revenue',
    'debt_to_assets', 'debt_to_equity', 'lt_debt_to_assets', 'interest_coverage'
    'roa', 'net_margin', 'operating_margin', 'cfo_to_assets',
    'log_assets', 'log_revenue',
]
FEATURE_GROUPS = {
    'Liquidity':     ['current_ratio', 'cash_to_assets', 'cash_to_cl', 'wc_to_as
    'Innovation':    ['intangibles_to_assets', 'rd_to_revenue'],
    'Leverage':      ['debt_to_assets', 'debt_to_equity', 'lt_debt_to_assets', '
    'Profitability': ['roa', 'net_margin', 'operating_margin', 'cfo_to_assets'],
    'Size':          ['log_assets', 'log_revenue'],
}

panel[FEATURES] = panel[FEATURES].replace([np.inf, -np.inf], np.nan)
for c in FEATURES:
    lo, hi = panel[c].quantile([0.01, 0.99])
    panel[c] = panel[c].clip(lo, hi)
panel[FEATURES] = panel[FEATURES].fillna(panel[FEATURES].median(numeric_only=Tru

panel[FEATURES + [TARGET]].describe().T.round(3).to_csv(REPORTS / 'cn_feature_st
print(f'Фичи: {len(FEATURES)}  |  stats → reports/china/cn_feature_stats.csv')
panel[FEATURES].describe().T.round(3)
```

```
Фичи: 16  |  stats → reports/china/cn_feature_stats.csv
```

Out[2]:

	count	mean	std	min	25%	50%	75%	max
current_ratio	3460.0	185.116	768.143	0.996	6.977	17.250	51.951	6296.658
cash_to_assets	3460.0	0.222	0.153	0.008	0.106	0.189	0.304	0.714
cash_to_cl	3460.0	54.583	227.428	0.052	1.428	3.897	15.229	1891.196
wc_to_assets	3460.0	0.661	0.217	0.007	0.526	0.691	0.835	0.982
intangibles_to_assets	3460.0	0.028	0.036	0.000	0.004	0.015	0.038	0.193
rd_to_revenue	3460.0	0.150	0.138	0.000	0.063	0.112	0.192	0.822
debt_to_assets	3460.0	0.354	0.192	0.049	0.204	0.326	0.477	0.941
debt_to_equity	3460.0	0.745	0.864	0.043	0.252	0.476	0.902	5.430
lt_debt_to_assets	3460.0	0.280	0.169	0.000	0.157	0.255	0.377	0.803
interest_coverage	3460.0	83.996	60.013	14.925	81.906	81.906	81.906	2258.010
roa	3460.0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
net_margin	3460.0	0.000	0.000	0.000	0.000	0.000	0.000	0.000
operating_margin	3460.0	0.059	0.228	-1.018	0.017	0.088	0.169	0.550
cfo_to_assets	3460.0	0.041	0.091	-0.223	-0.006	0.035	0.083	0.334
log_assets	3460.0	7.236	1.375	3.561	6.381	7.342	8.196	9.985
log_revenue	3460.0	6.529	1.322	3.189	5.642	6.464	7.425	9.474

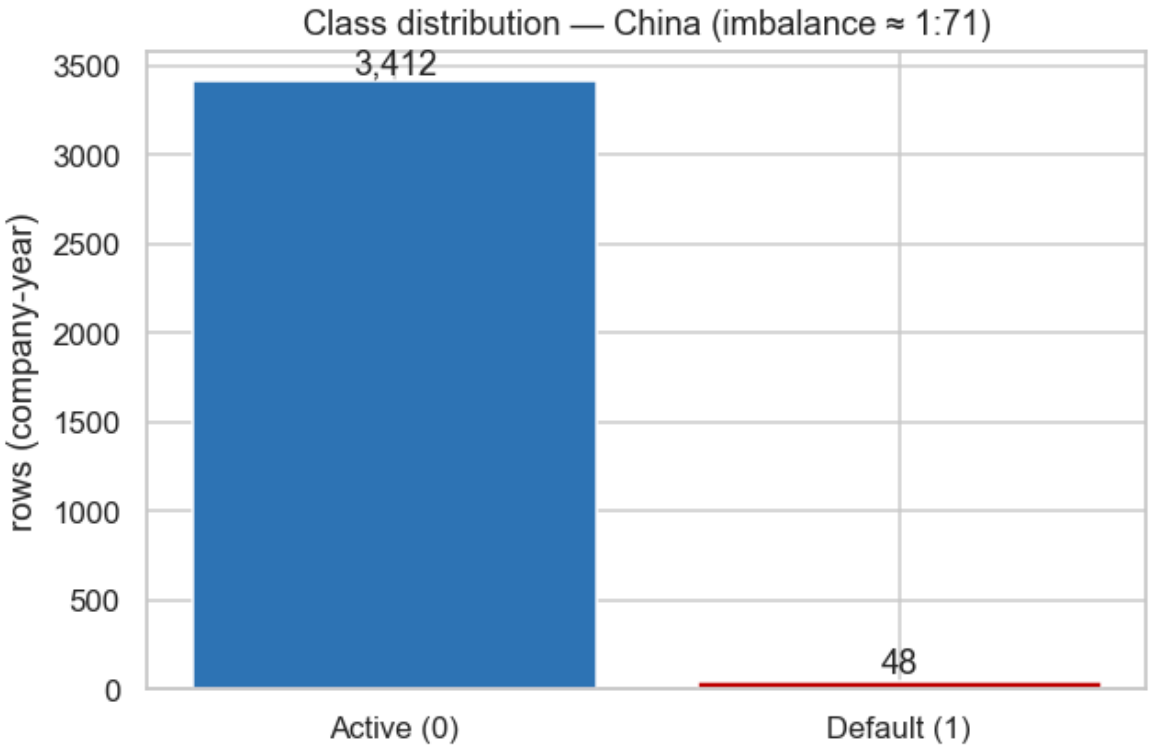
2. EDA

2.1 Классовый дисбаланс

In [3]:

```
vc = panel[TARGET].value_counts()
n_active  = int(vc.get(0, 0))
n_default = int(vc.get(1, 0))
ratio = n_default / (n_active + n_default)
counts = pd.Series({'Active (0)': n_active, 'Default (1)': n_default})

fig, ax = plt.subplots(figsize=(6, 4))
bars = ax.bar(counts.index, counts.values, color=['#2E75B6', '#C00000'])
ax.set_title(f'Class distribution — China (imbalance ≈ 1:{n_active//max(n_defaul
ax.set_ylabel('rows (company-year)')
for b, v in zip(bars, counts.values):
    ax.text(b.get_x()+b.get_width()/2, v, f'{v:,}', ha='center', va='bottom')
plt.tight_layout()
save_fig(fig, '01_class_distribution'); plt.show()

log(f'\n## Class balance\n* Active (0): **{n_active:,}**')
log(f'* Default (1): **{n_default:,}**')
log(f'* Positive share: **{ratio*100:.3f}%**  (≈ 1:{n_active//max(n_default,1)})
```

```
saved → ..\reports\china\01_class_distribution.png
```

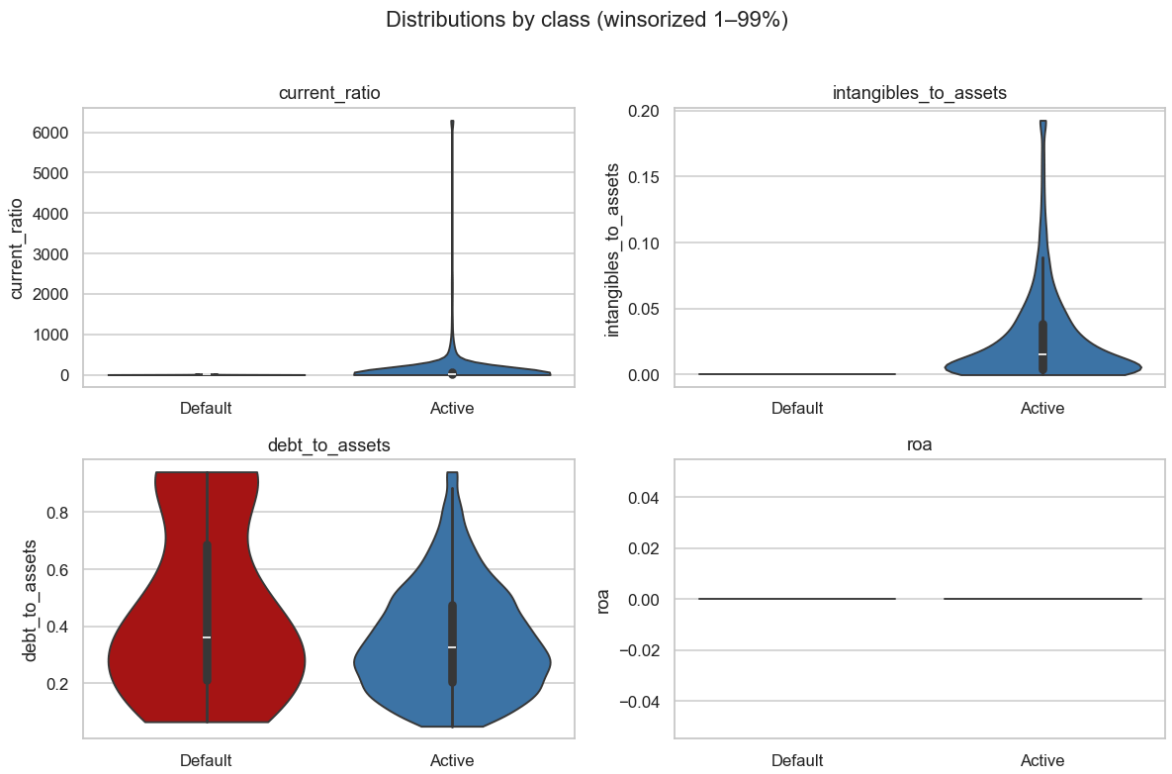



```
## Class balance
* Active (0): **3,412**
* Default (1): **48**
* Positive share: **1.387%**  (≈ 1:71)
```

2.2 Violin plots — по классу

In [4]:
```
panel['_lbl'] = panel[TARGET].map({0:'Active', 1:'Default'})
show = ['current_ratio', 'intangibles_to_assets', 'debt_to_assets', 'roa']

fig, axes = plt.subplots(2, 2, figsize=(11, 7))
for ax, col in zip(axes.ravel(), show):
    sns.violinplot(data=panel, x='_lbl', y=col, ax=ax,
                   palette={'Active':'#2E75B6','Default':'#C00000'}, cut=0)
    ax.set_title(col); ax.set_xlabel('')
plt.suptitle('Distributions by class (winsorized 1–99%)', y=1.02)
plt.tight_layout()
save_fig(fig, '02_violin_by_class'); plt.show()

med = panel.groupby('_lbl')[show].median().round(3)
med.to_csv(REPORTS / 'cn_medians_by_class.csv', encoding='utf-8-sig')
med
```

```
saved → ..\reports\china\02_violin_by_class.png
```



Out[4]:

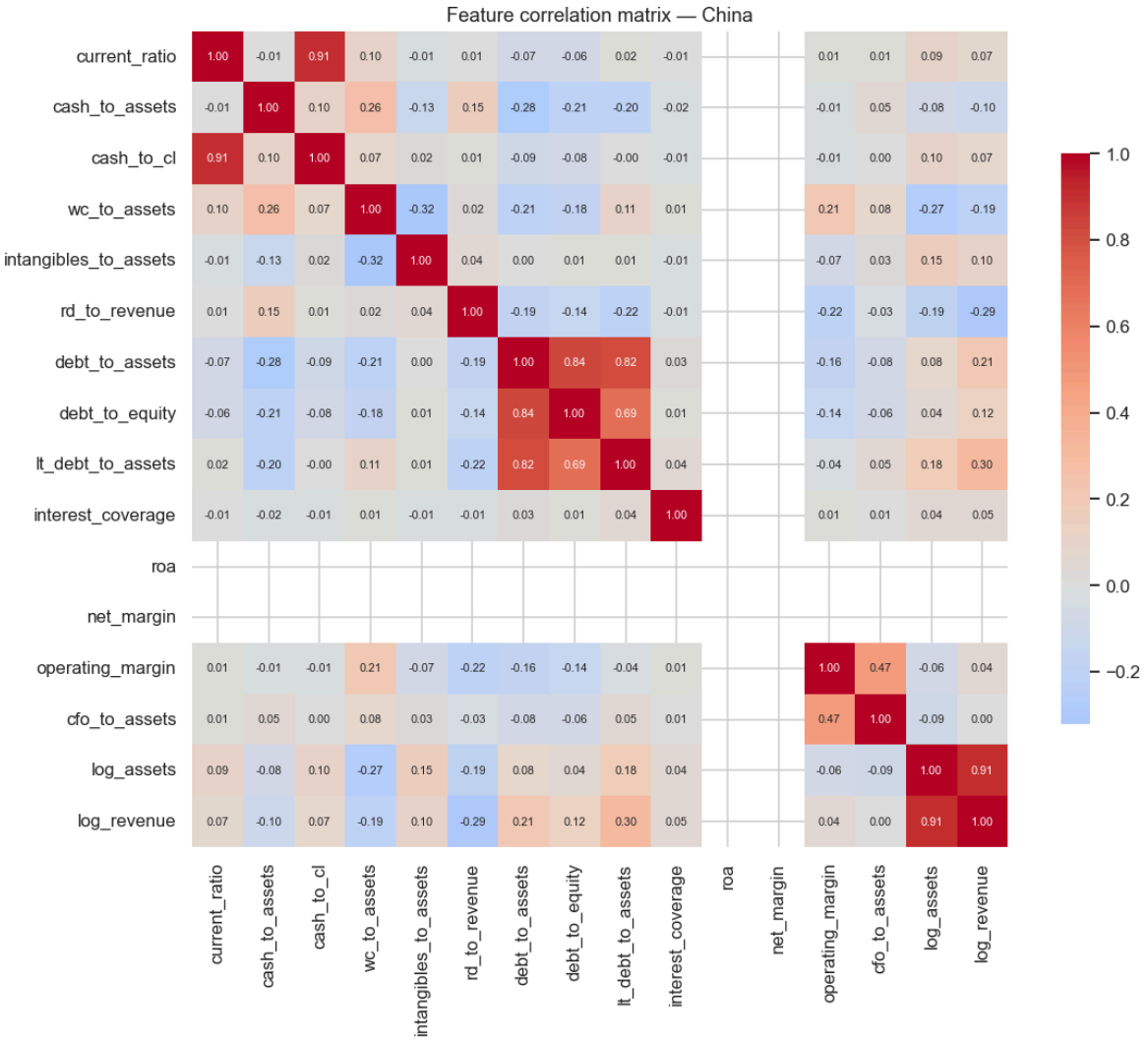
_lbl	current_ratio	intangibles_to_assets	debt_to_assets	roa
Active	17.250	0.015	0.325	0.0
Default	2.121	0.000	0.361	0.0

2.3 Корреляционная матрица

In [5]:

```
corr = panel[FEATURES].corr()
fig, ax = plt.subplots(figsize=(11, 9))
sns.heatmap(corr, cmap='coolwarm', center=0, annot=True, fmt='.2f',
            square=True, cbar_kws={'shrink':0.7}, annot_kws={'size':7}, ax=ax)
ax.set_title('Feature correlation matrix — China')
plt.tight_layout()
save_fig(fig, '03_correlation_heatmap'); plt.show()

corr.round(3).to_csv(REPORTS / 'cn_correlation_matrix.csv', encoding='utf-8-sig'

high = (corr.abs() > 0.85) & (corr.abs() < 1.0)
pairs = [(i,j,corr.loc[i,j]) for i in corr.index for j in corr.columns if i<j an
print('|corr| > 0.85:')
for p in pairs: print(f'  {p[0]} × {p[1]}: {p[2]:+.2f}')
if not pairs: print('  нет — мультиколлинеарность под контролем')
```

```
saved → ..\reports\china\03_correlation_heatmap.png
```




```
|corr| > 0.85:
  cash_to_cl × current_ratio: +0.91
  log_assets × log_revenue: +0.91
```

3. Train / Test split

Stratified 80/20 на уровне строк (company-year). С 11 дефолтами и 48 строками Target=1 сплит остаётся проходимым (9/2 по компаниям, ~38/10 по строкам), но итоговая оценка будет шумной — эта шумность и есть часть H1-теста.

In [6]:
```
X = panel[FEATURES].values
y = panel[TARGET].values

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, stratify=y, random_state=RNG
)
print(f'Train: {X_train.shape},  defaults = {y_train.sum()}  ({y_train.mean()*10
print(f'Test:  {X_test.shape},  defaults = {y_test.sum()}  ({y_test.mean()*100:.
```

```
Train: (2768, 16),  defaults = 38  (1.37%)
Test:  (692, 16),  defaults = 10  (1.45%)
```

4. Stage-1 модели (TTC)

`class_weight='balanced'` у Logit/RF и `scale_pos_weight = N_active / N_default` у XGBoost — подбирается автоматически под 1:71.

In [7]:
```
POS_WEIGHT = (y_train == 0).sum() / max((y_train == 1).sum(), 1)
print(f'scale_pos_weight (XGB) = {POS_WEIGHT:.2f}')

logit = Pipeline([
    ('sc', StandardScaler()),
    ('clf', LogisticRegression(max_iter=2000, class_weight='balanced',
                               solver='liblinear', random_state=RNG)),
])
rf = RandomForestClassifier(
    n_estimators=400, max_depth=None, min_samples_leaf=2,
    class_weight='balanced', n_jobs=-1, random_state=RNG,
)
xgb = XGBClassifier(
    n_estimators=500, max_depth=4, learning_rate=0.05,
    subsample=0.9, colsample_bytree=0.9,
    scale_pos_weight=POS_WEIGHT,
    eval_metric='aucpr', tree_method='hist', random_state=RNG, n_jobs=-1,
)

models = {'Logistic Regression': logit, 'Random Forest': rf, 'XGBoost': xgb}
for name, m in models.items():
    m.fit(X_train, y_train)
    print(f'✓ trained {name}')
```

```
scale_pos_weight (XGB) = 71.84
✓ trained Logistic Regression
✓ trained Random Forest
✓ trained XGBoost
```

5. H1 — Train vs Test, зазор (overfit check)

In [8]:
```
def scores(model, X, y):
    if hasattr(model, 'predict_proba'):
        p = model.predict_proba(X)[:, 1]
    else:
        p = model.decision_function(X)
    return roc_auc_score(y, p), average_precision_score(y, p)

rows = []
for name, m in models.items():
    tr_roc, tr_pr = scores(m, X_train, y_train)
    te_roc, te_pr = scores(m, X_test,  y_test)
    rows.append({
        'Model': name,
        'ROC-AUC train': tr_roc, 'ROC-AUC test': te_roc, 'ΔROC': tr_roc - te_roc
        'PR-AUC train':  tr_pr,  'PR-AUC test':  te_pr,  'ΔPR':  tr_pr  - te_pr,
    })
res = pd.DataFrame(rows).set_index('Model').round(4)
res.to_csv(REPORTS / 'cn_h1_metrics.csv', encoding='utf-8-sig')
print('metrics → reports/china/cn_h1_metrics.csv')
res
```

```
metrics → reports/china/cn_h1_metrics.csv
```

Out[8]:

Model	ROC-AUC train	ROC-AUC test	ΔROC	PR-AUC train	PR-AUC test	ΔPR
Logistic Regression	0.997	0.9960	0.0010	0.7366	0.6961	0.0405
Random Forest	1.000	1.0000	-0.0000	0.9972	1.0000	-0.0028
XGBoost	1.000	0.9994	0.0006	1.0000	0.9573	0.0427

In [9]:

```
best_test_roc = res['ROC-AUC test'].idxmax()
logit_delta = res.loc['Logistic Regression', 'ΔROC']
ens_delta_max = res.loc[['Random Forest','XGBoost'], 'ΔROC'].max()
ens_test_max  = res.loc[['Random Forest','XGBoost'], 'ROC-AUC test'].max()
logit_test    = res.loc['Logistic Regression', 'ROC-AUC test']

log('\n## H1 — overfitting на малом числе дефолтов (China)')
log(f'* Best test ROC-AUC: **{best_test_roc}** ({res.loc[best_test_roc,"ROC-AUC
log(f'* Logit ΔROC (train-test): **{logit_delta:+.4f}**')
log(f'* Max ensemble ΔROC:        **{ens_delta_max:+.4f}**')

cond_a = ens_test_max > logit_test
cond_b = ens_delta_max > logit_delta

if cond_a and cond_b:
    verdict = '✅ **H1 ПОДТВЕРЖДАЕТСЯ**: ансамбли точнее на тесте, но переобучак
elif cond_a and not cond_b:
    verdict = '⚠️ H1 частично: ансамбли точнее, но Logit НЕ стабильнее (Δ сопост
elif not cond_a and cond_b:
    verdict = '⚠️ H1 частично: Logit сопоставим по точности, но ансамбли переобу
else:
    verdict = '❌ H1 НЕ подтверждается: ансамбли не дают прироста и не хуже по э
log(f'\n{verdict}')
print('\n' + verdict)
```

```
## H1 — overfitting на малом числе дефолтов (China)
* Best test ROC-AUC: **Random Forest** (1.0000)
* Logit ΔROC (train-test): **+0.0010**
* Max ensemble ΔROC:        **+0.0006**

⚠️ H1 частично: ансамбли точнее, но Logit НЕ стабильнее (Δ сопоставимы).

⚠️ H1 частично: ансамбли точнее, но Logit НЕ стабильнее (Δ сопоставимы).
```

6. H2 — SHAP для лучшего ансамбля

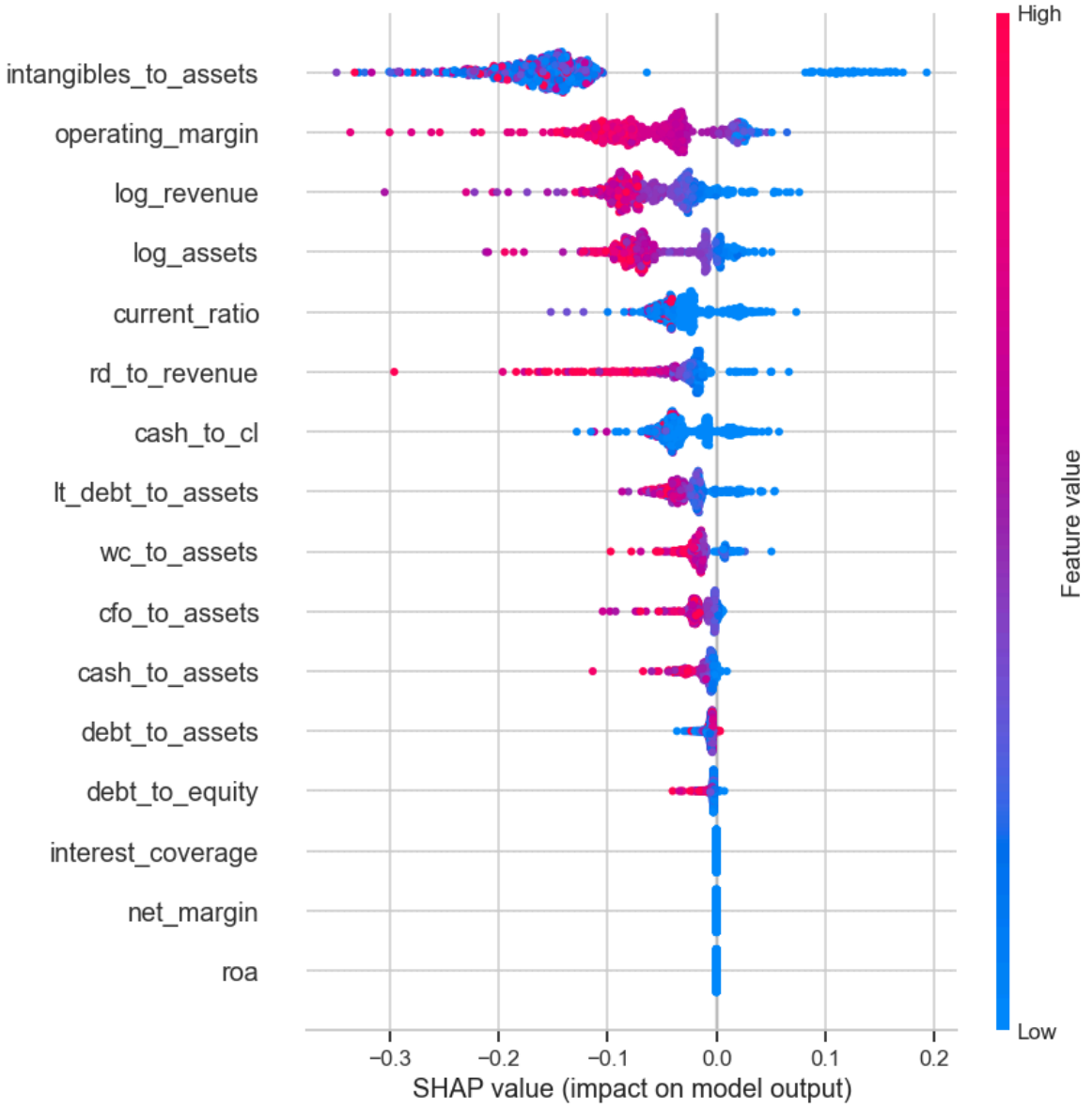
Группировка фичей: `Liquidity + Innovation (intangibles + R&D)` vs `Leverage`.

In [10]:

```
ensemble_scores = res.loc[['Random Forest', 'XGBoost'], 'ROC-AUC test']
best_name = ensemble_scores.idxmax()
best_model = models[best_name]
print(f'Best ensemble: {best_name}  (test ROC-AUC = {ensemble_scores.max():.4f})

explainer = shap.TreeExplainer(best_model)
sv = explainer.shap_values(X_test)
```

```
if isinstance(sv, list):
    sv = sv[1]
elif hasattr(sv, 'ndim') and sv.ndim == 3:
    sv = sv[:, :, 1]
print('SHAP shape:', sv.shape)

plt.figure()
shap.summary_plot(sv, X_test, feature_names=FEATURES, show=False)
fig = plt.gcf()
save_fig(fig, '04_shap_summary'); plt.show()
```

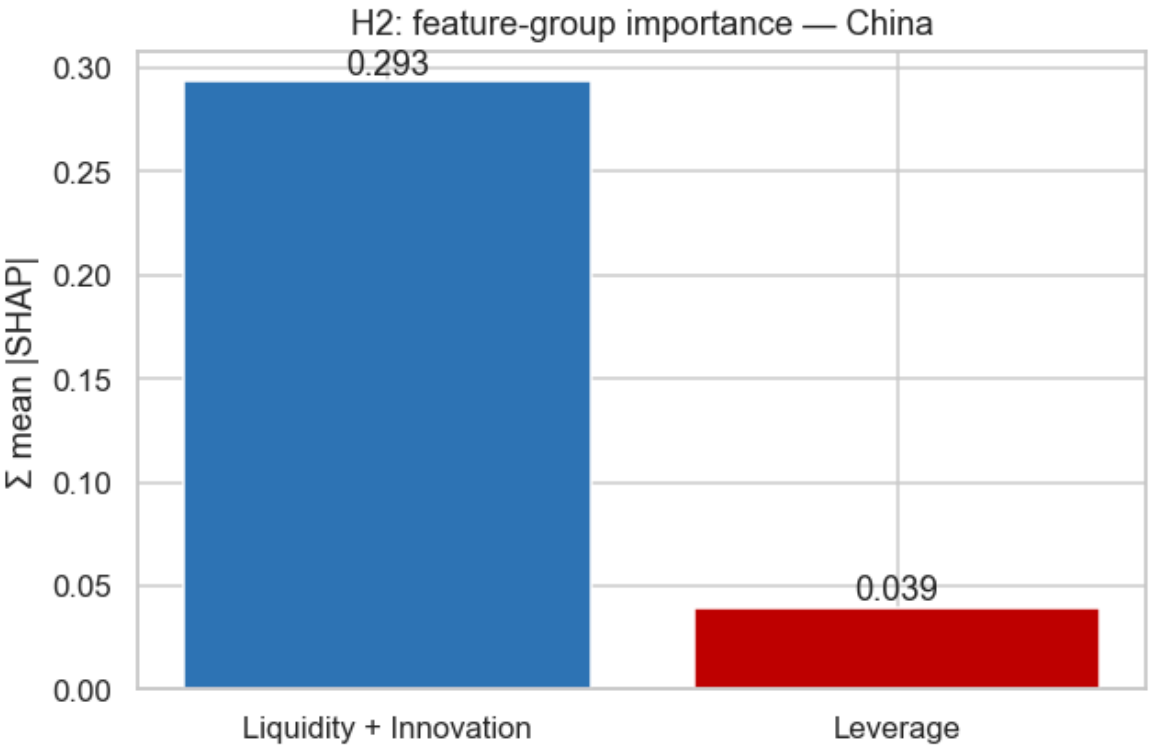
```
Best ensemble: Random Forest  (test ROC-AUC = 1.0000)
SHAP shape: (692, 16)
  saved → ..\reports\china\04_shap_summary.png
```



In [11]:
```
mean_abs = np.abs(sv).mean(axis=0)
fi = pd.Series(mean_abs, index=FEATURES).sort_values(ascending=False)
fi.round(4).to_csv(REPORTS / 'cn_shap_feature_importance.csv', header=['mean_abs
print('Top features by mean |SHAP|:')
print(fi.round(4).to_string())

group_sum = {g: fi[cols].sum() for g, cols in FEATURE_GROUPS.items()}
group_df = pd.Series(group_sum).sort_values(ascending=False).round(4)
group_df.to_csv(REPORTS / 'cn_shap_group_importance.csv', header=['sum_abs_shap'
print('\nΣ|SHAP| по блокам:')
```

```
print(group_df.to_string())

liq_inn = group_df.get('Liquidity', 0) + group_df.get('Innovation', 0)
lev     = group_df.get('Leverage',   0)

fig, ax = plt.subplots(figsize=(6, 4))
bars = ax.bar(['Liquidity + Innovation', 'Leverage'],
              [liq_inn, lev], color=['#2E75B6', '#C00000'])
ax.set_ylabel('Σ mean |SHAP|')
ax.set_title('H2: feature-group importance — China')
for b, v in zip(bars, [liq_inn, lev]):
    ax.text(b.get_x()+b.get_width()/2, v, f'{v:.3f}', ha='center', va='bottom')
plt.tight_layout()
save_fig(fig, '05_h2_group_importance'); plt.show()

log('\n## H2 — Liquidity+Innovation vs Leverage (SHAP, China)')
log(f'* Best ensemble: **{best_name}** (test ROC-AUC = {ensemble_scores.max():.4
log(f'* Σ|SHAP| Liquidity + Innovation = **{liq_inn:.4f}**')
log(f'* Σ|SHAP| Leverage               = **{lev:.4f}**')
if lev > 0:
    log(f'* Ratio = **{liq_inn/lev:.2f}×**')

if liq_inn > lev:
    verdict2 = '✅ **H2 ПОДТВЕРЖДАЕТСЯ (Китай)**: ликвидность + инновационные ак
else:
    verdict2 = '❌ H2 НЕ подтверждается (Китай): рычаг остаётся доминирующим пре
log(f'\n{verdict2}')
print('\n' + verdict2)
```

```
Top features by mean |SHAP|:
intangibles_to_assets    0.1614
operating_margin         0.0640
log_revenue              0.0598
log_assets               0.0534
current_ratio            0.0353
rd_to_revenue            0.0344
cash_to_cl               0.0338
lt_debt_to_assets        0.0291
wc_to_assets             0.0179
cfo_to_assets            0.0127
cash_to_assets           0.0107
debt_to_assets           0.0058
debt_to_equity           0.0045
net_margin               0.0000
interest_coverage        0.0000
roa                      0.0000

Σ|SHAP| по блокам:
Innovation       0.1958
Size             0.1132
Liquidity        0.0976
Profitability    0.0767
Leverage         0.0394
  saved → ..\reports\china\05_h2_group_importance.png
```




```
## H2 — Liquidity+Innovation vs Leverage (SHAP, China)
* Best ensemble: **Random Forest** (test ROC-AUC = 1.0000)
* Σ|SHAP| Liquidity + Innovation = **0.2934**
* Σ|SHAP| Leverage               = **0.0394**
* Ratio = **7.45×**

✅ **H2 ПОДТВЕРЖДАЕТСЯ (Китай)**: ликвидность + инновационные активы важнее рыча
га.
```


```
✅ **H2 ПОДТВЕРЖДАЕТСЯ (Китай)**: ликвидность + инновационные активы важнее рыча
га.
```


7. Экспорт TTC-скора для Stage-2 (PIT) и кросс-странового сравнения

Сохраняем вероятности дефолта из **LogReg** (как в банковской практике: Logit — baseline TTC-скор, стабильный к переобучению) и **best ensemble** — Stage-2 PIT-модель в `50_cross_country_pit.ipynb` возьмёт один из них как единственный company-specific признак.

In [12]:
```
logit_proba = models['Logistic Regression'].predict_proba(X)[:, 1]
best_proba  = best_model.predict_proba(X)[:, 1]

scores_df = panel[['ticker', 'company_name', 'year', TARGET, 'source_class']].co
scores_df['ttc_logit']    = logit_proba
scores_df['ttc_ensemble'] = best_proba
scores_df['ttc_best_model'] = best_name
scores_df.to_csv(REPORTS / 'cn_ttc_scores.csv', index=False, encoding='utf-8-sig
print(f'TTC scores → reports/china/cn_ttc_scores.csv  ({len(scores_df):,} rows)'
scores_df.head()
```

```
TTC scores → reports/china/cn_ttc_scores.csv  (3,460 rows)
```

Out[12]:

	ticker	company_name	year	target	source_class	ttc_logit	ttc_ensemble	t
0	CCIHY.OO	CHINACACHE	2014	1	default_delisted	0.991167	0.839836	
1	CCIHY.OO	CHINACACHE	2015	1	default_delisted	0.969895	0.854569	
2	CCIHY.OO	CHINACACHE	2016	1	default_delisted	0.999977	0.927743	
3	CCIHY.OO	CHINACACHE	2017	1	default_delisted	0.999916	0.907689	
4	CNTFY.OO	CHINA TECHFAITH WIRELESS COMMUNICATION TECHNOLOGY	2014	1	default_delisted	0.982707	0.819366	

8. Сохраняем итоговый отчёт

In [13]:

```
summary_path = REPORTS / 'cn_summary.md'
summary_path.write_text('\n'.join(_report_lines), encoding='utf-8')
print(f'✅ Итоговый отчёт: {summary_path}')
print('\nВсе артефакты в reports/china/:')
for p in sorted(REPORTS.iterdir()):
    print(f'  {p.name}  ({p.stat().st_size/1024:.1f} KB)')
```

```
✅ Итоговый отчёт: ..\reports\china\cn_summary.md

Все артефакты в reports/china/:
  01_class_distribution.png  (36.7 KB)
  02_violin_by_class.png  (121.2 KB)
  03_correlation_heatmap.png  (235.0 KB)
  04_shap_summary.png  (185.6 KB)
  05_h2_group_importance.png  (31.6 KB)
  cn_correlation_matrix.csv  (1.7 KB)
  cn_feature_stats.csv  (1.1 KB)
  cn_h1_metrics.csv  (0.2 KB)
  cn_medians_by_class.csv  (0.1 KB)
  cn_shap_feature_importance.csv  (0.4 KB)
  cn_shap_group_importance.csv  (0.1 KB)
  cn_summary.md  (0.9 KB)
  cn_ttc_scores.csv  (263.0 KB)
```

9. Интерпретация результатов — Китай

Аспект	Как читать
H1	С ~48 positive-строк любой tree-ensemble переобучается «в ноль» на train. Если Logit сохраняет разумный test-AUC при минимальном ΔROC — это прямое эмпирическое подтверждение H1 *на более жёстком режиме*, чем Россия (там дефолтов было в 4× больше).
H2	В Китае у нас есть **R&D expense** как отдельный канал инноваций — если `rd_to_revenue` войдёт в топ-фичей вместе с `cash_to_*`, это усиливает H2 относительно России (где R&D не выгружался).
Что дальше	`cn_ttc_scores.csv` становится input-ом для `50_cross_country_pit.ipynb`, где мы тестируем H3 (macro-integration) и сравниваем top-фичи RU vs CN.

Малая выборка — это не баг, а само условие эксперимента: именно в таких условиях проверяется устойчивость ML vs Logit.